Flows and Chaos

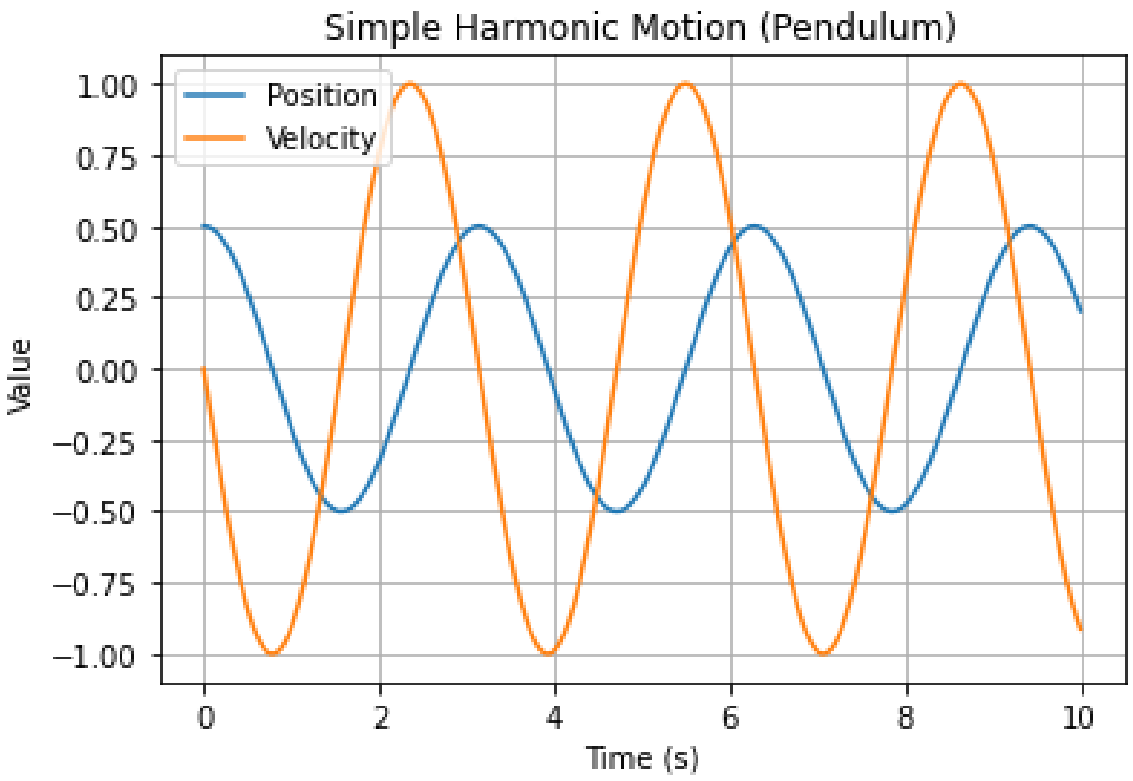
A higher order differential equation can be written as a set of first order differential equations.

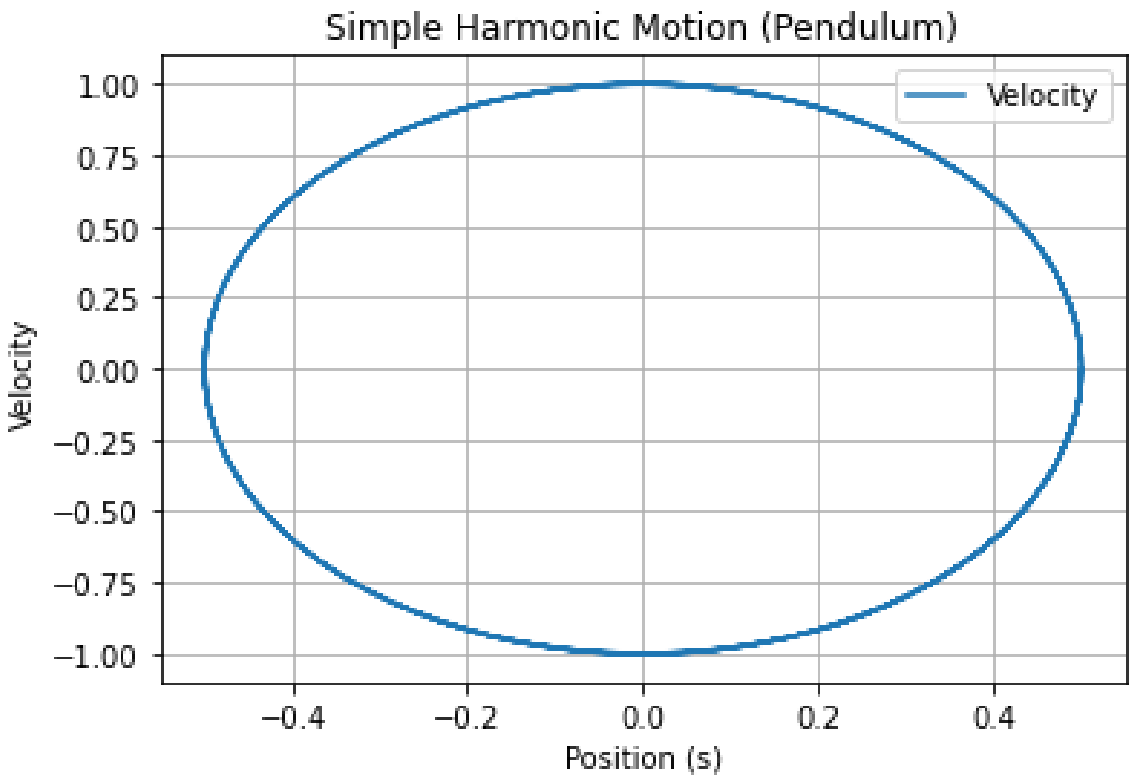
$$\ddot{x} = -\omega^2 x.$$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\omega^2 x_1 \end{aligned}$$

Flows and Chaos

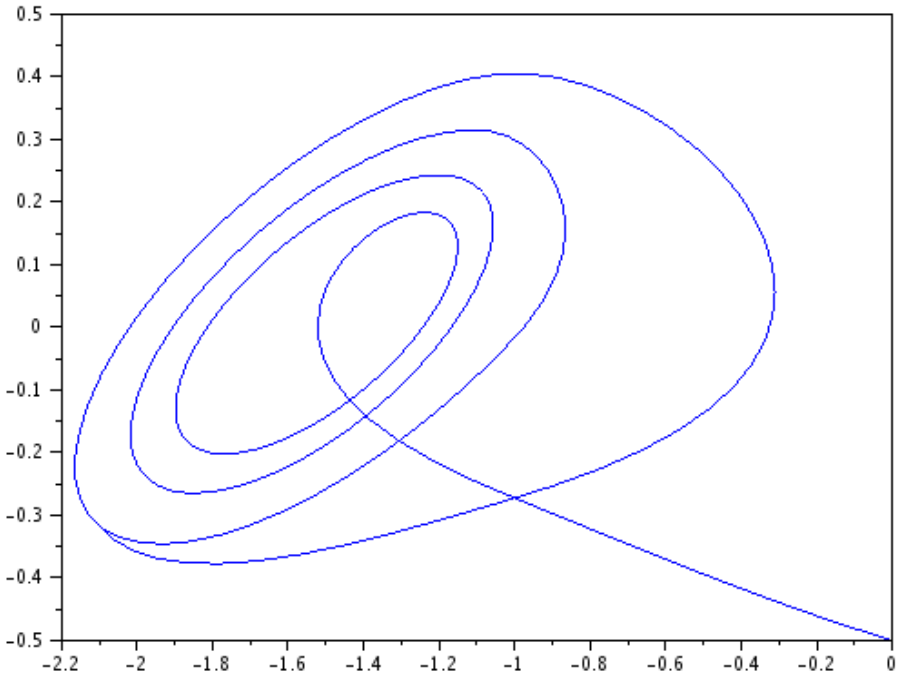
Then they can be solved using a solver in Python, to draw the trajectory in phase space, or the time variation of phase space variables.





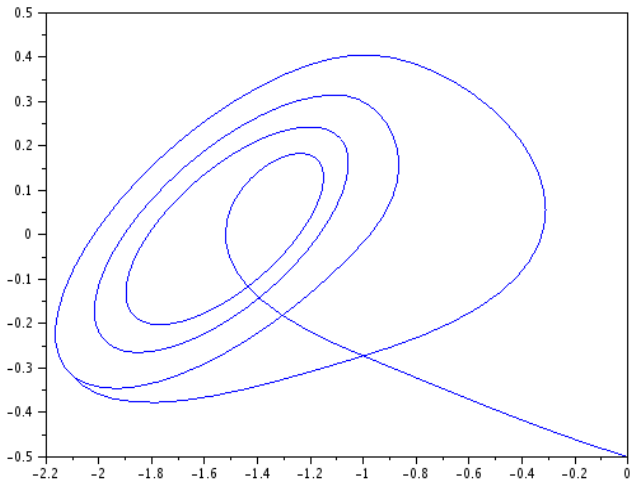
Flows and Chaos

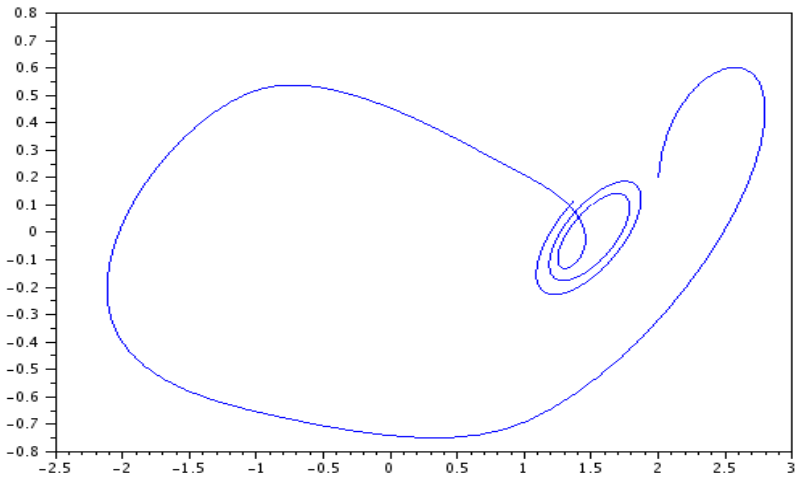
In order to solve a differential equation, we have to give an initial condition.



Flows and Chaos

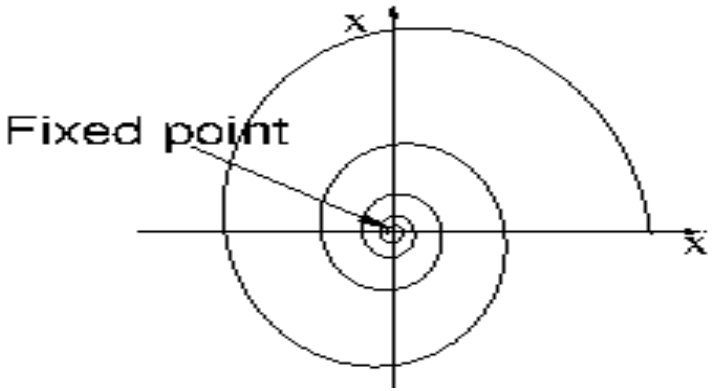
In general, different initial conditions give different trajectories.

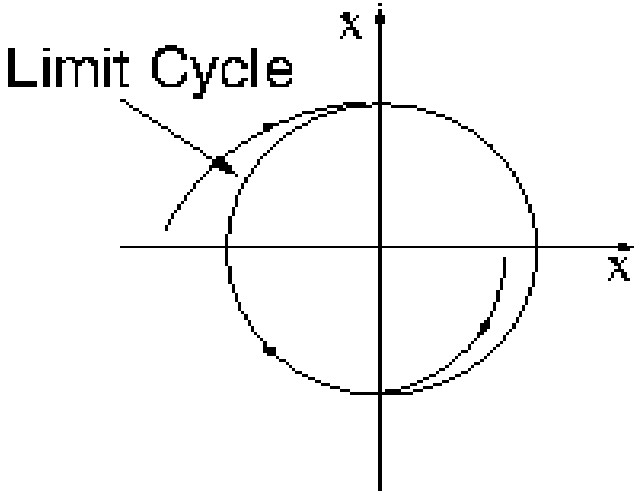




Flows and Chaos

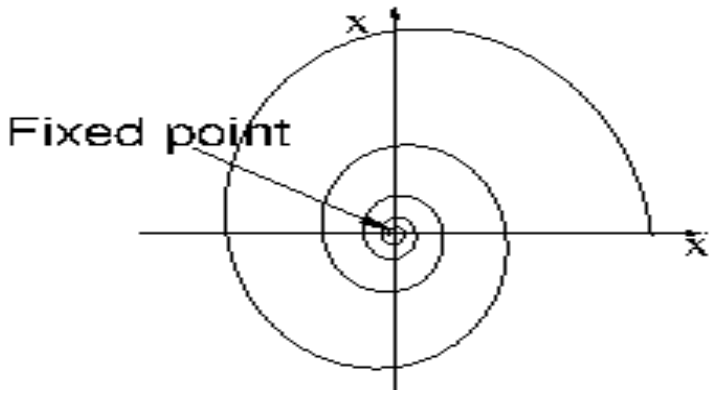
However, if there is a simple attractor like a fixed-point attractor or a limit cycle, many different initial conditions will give the same long-term behaviour.



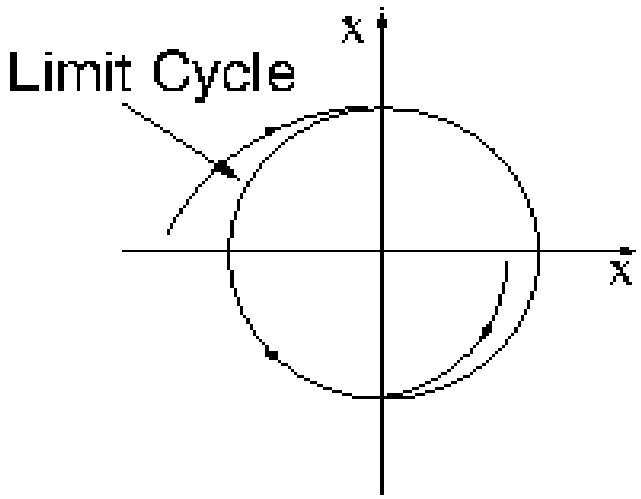


Flows and Chaos

If there is a fixed-point attractor in a system, it is an equilibrium point of that system.

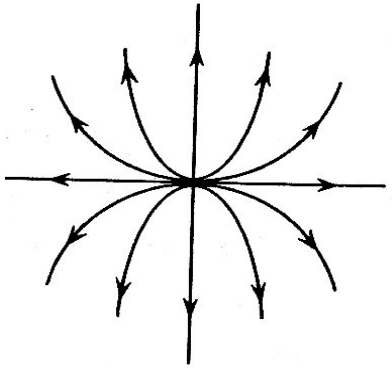


A limit cycle attractor is a periodic solution.

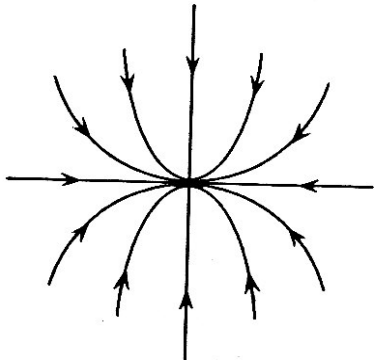


Flows and Chaos

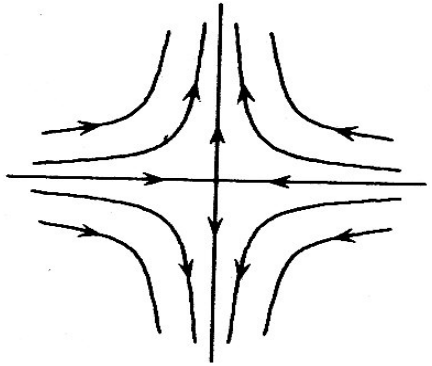
In addition to attractors, some systems can have unstable equilibrium points or unstable periodic solution.



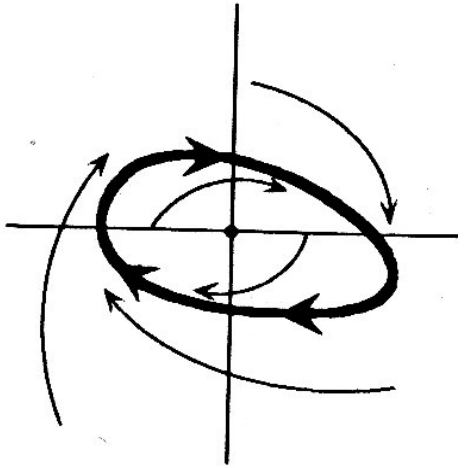
source



sink

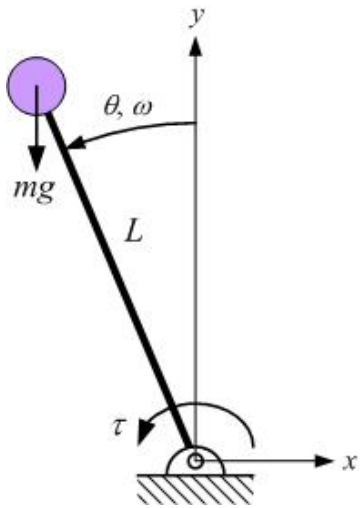


saddle node



Limit cycle

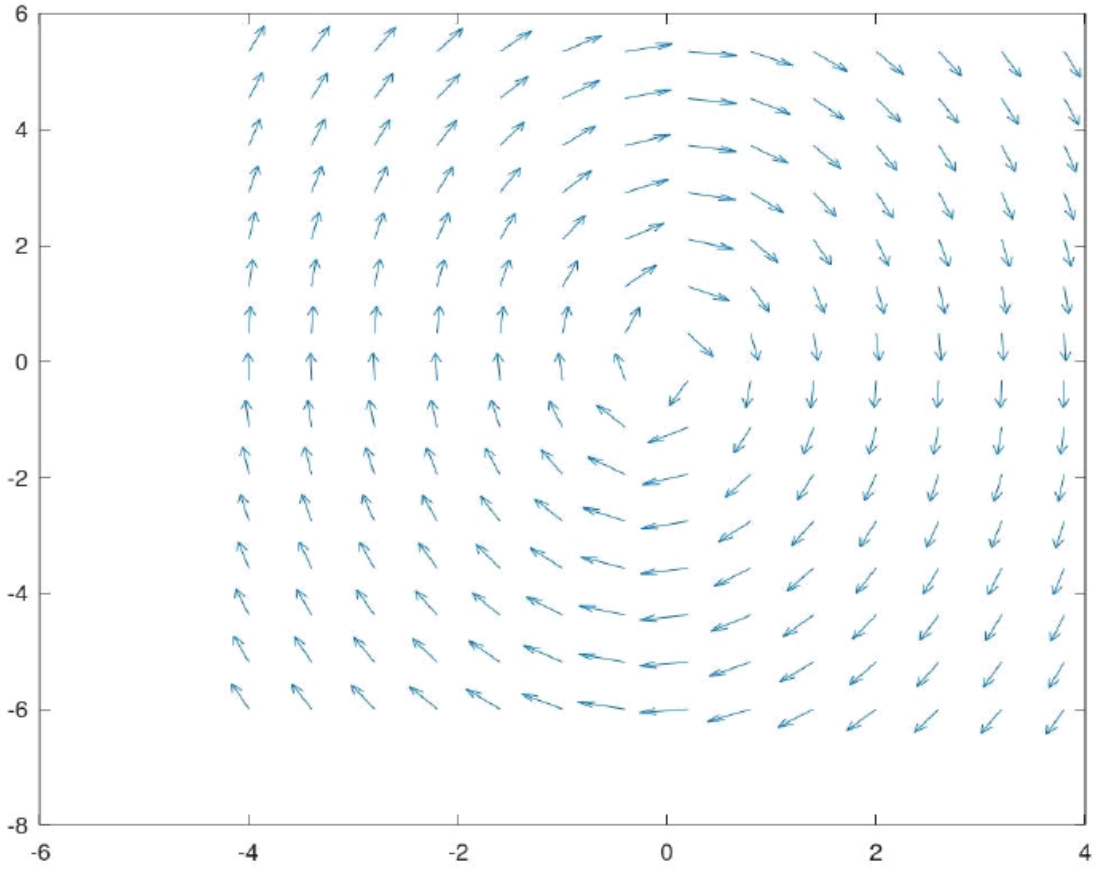
Example: the highest point of a pendulum (180-degree swing) with a zero velocity is an unstable equilibrium point.



Inverted pendulum

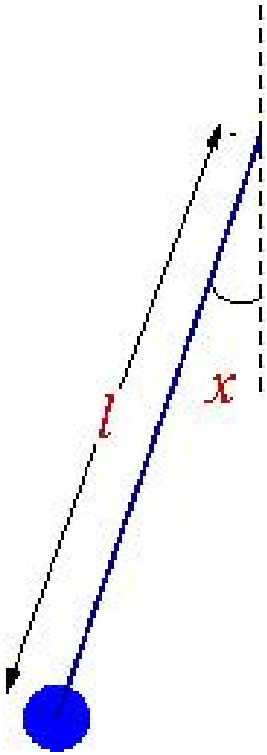
Last week

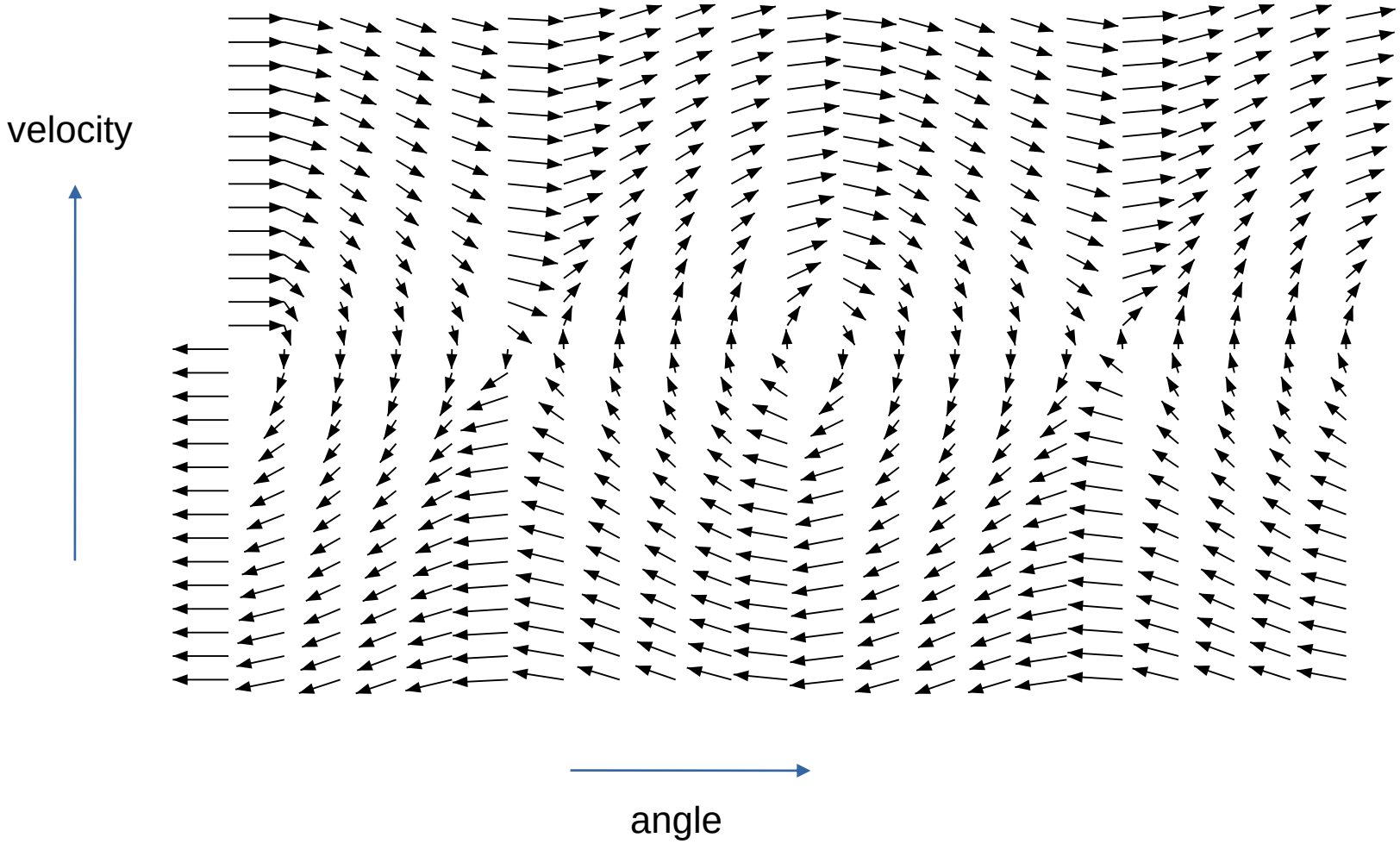
Flow diagram

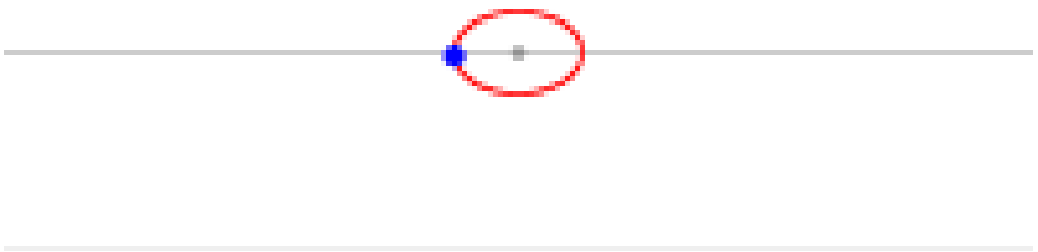


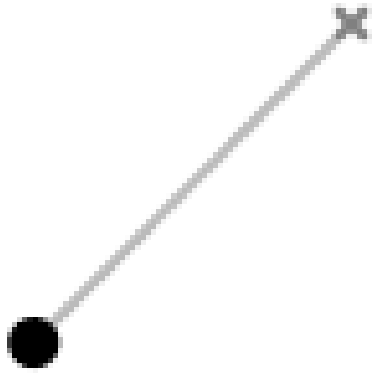
Last week

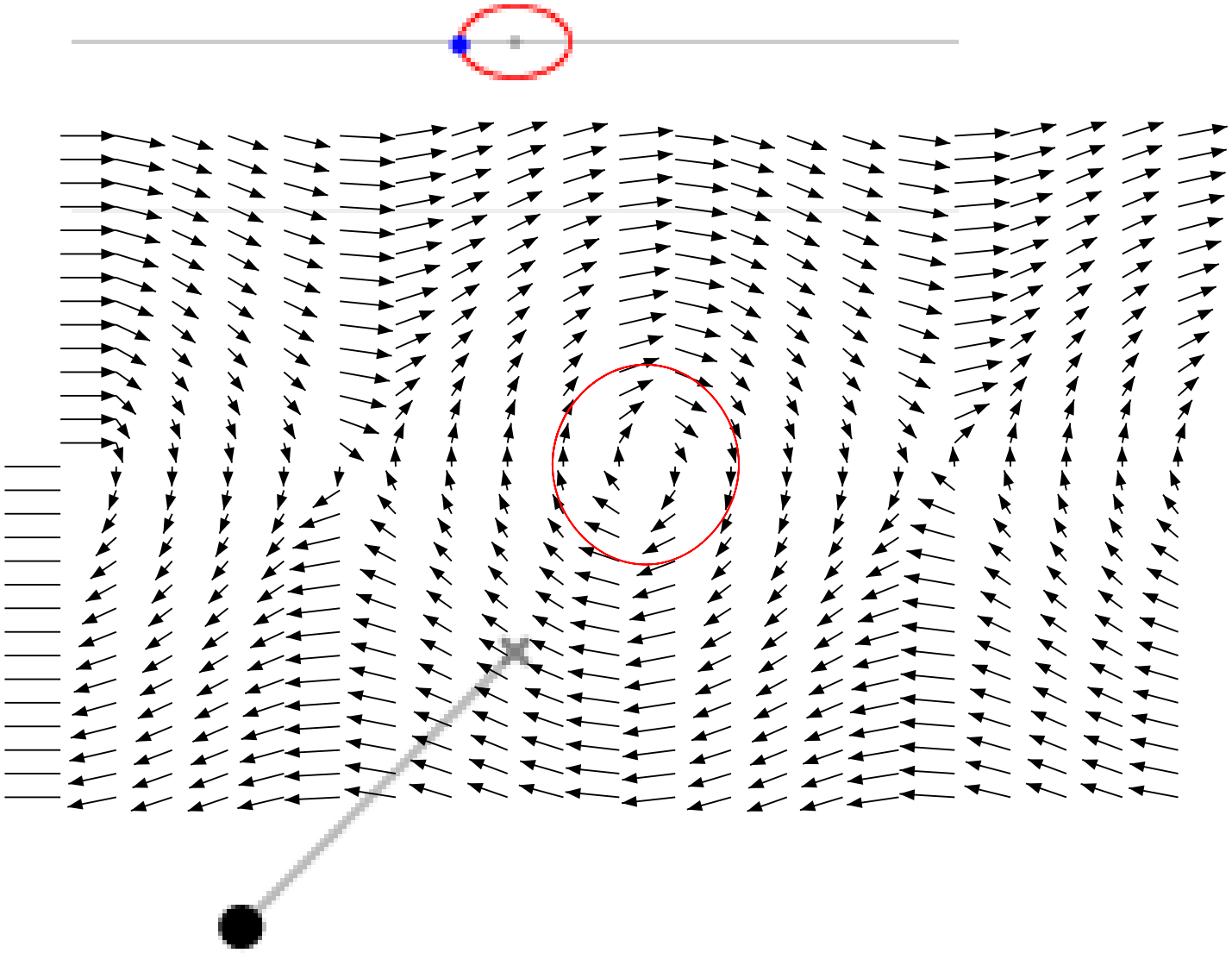
Pendulum

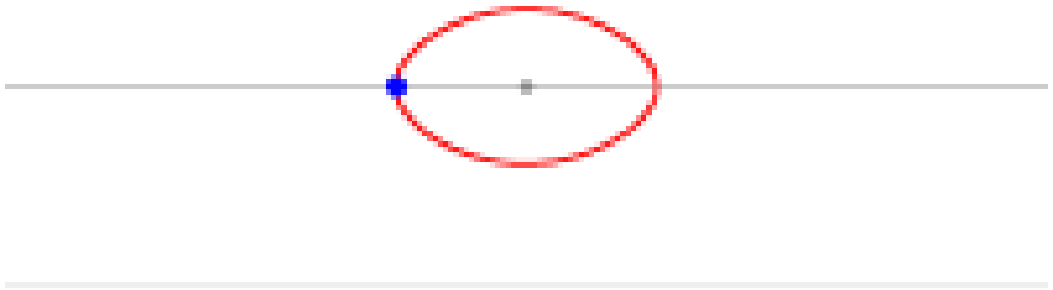


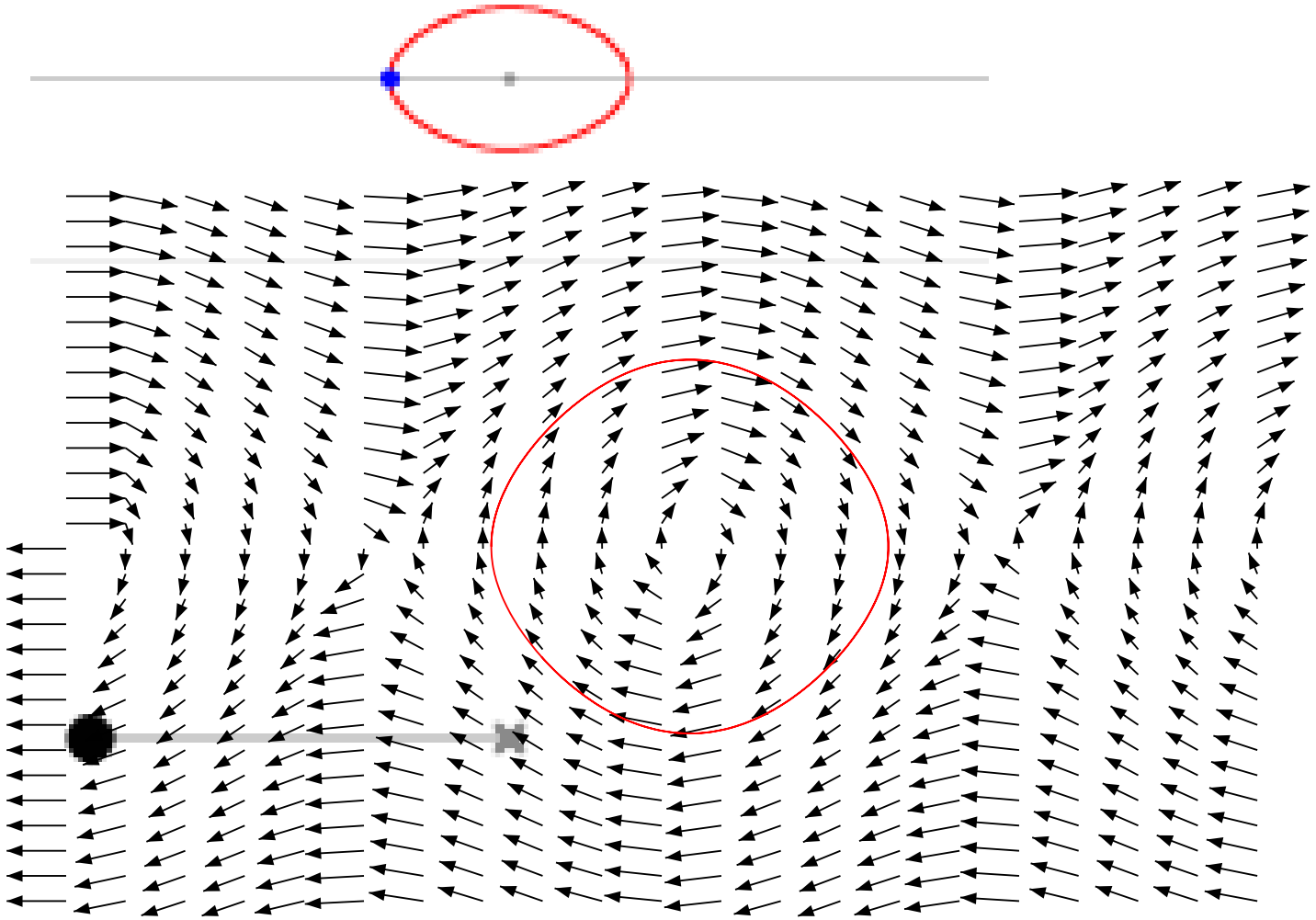

velocity
angle

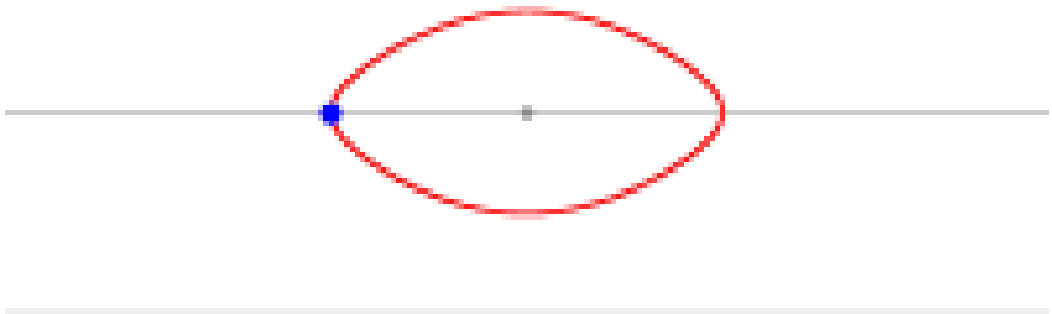


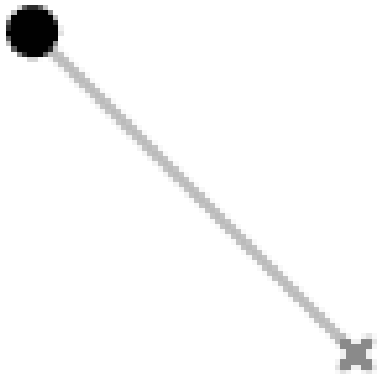


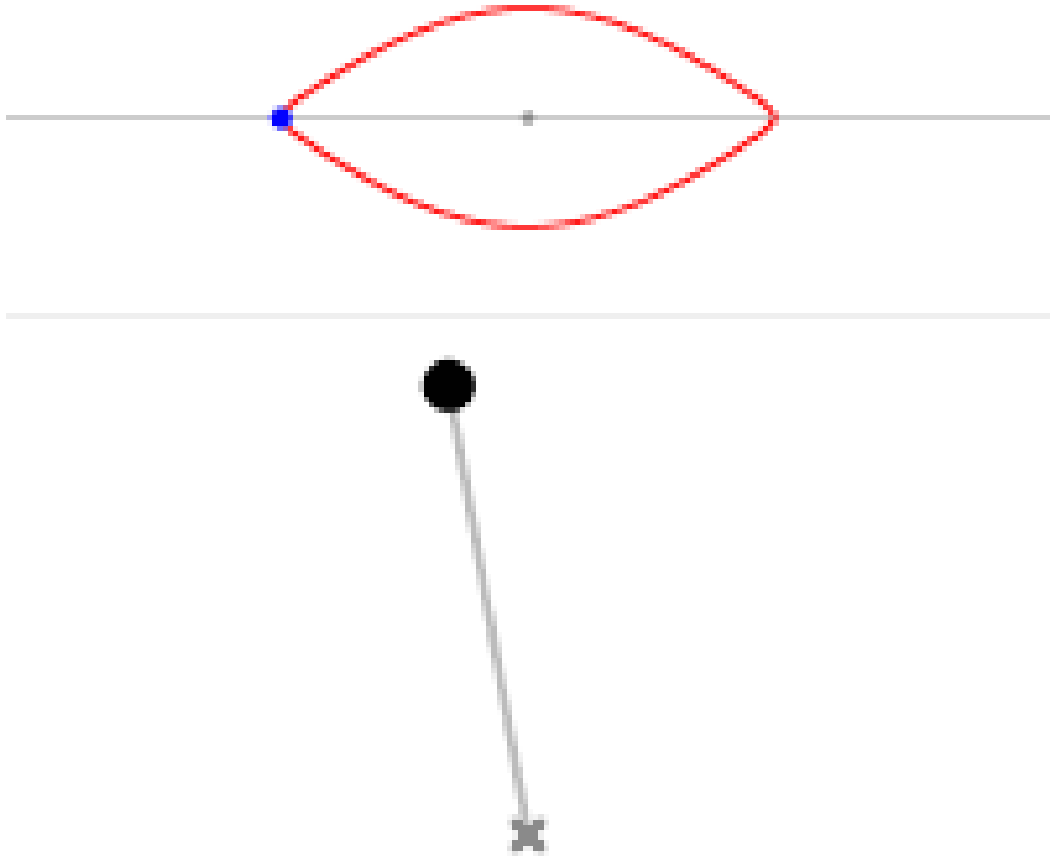


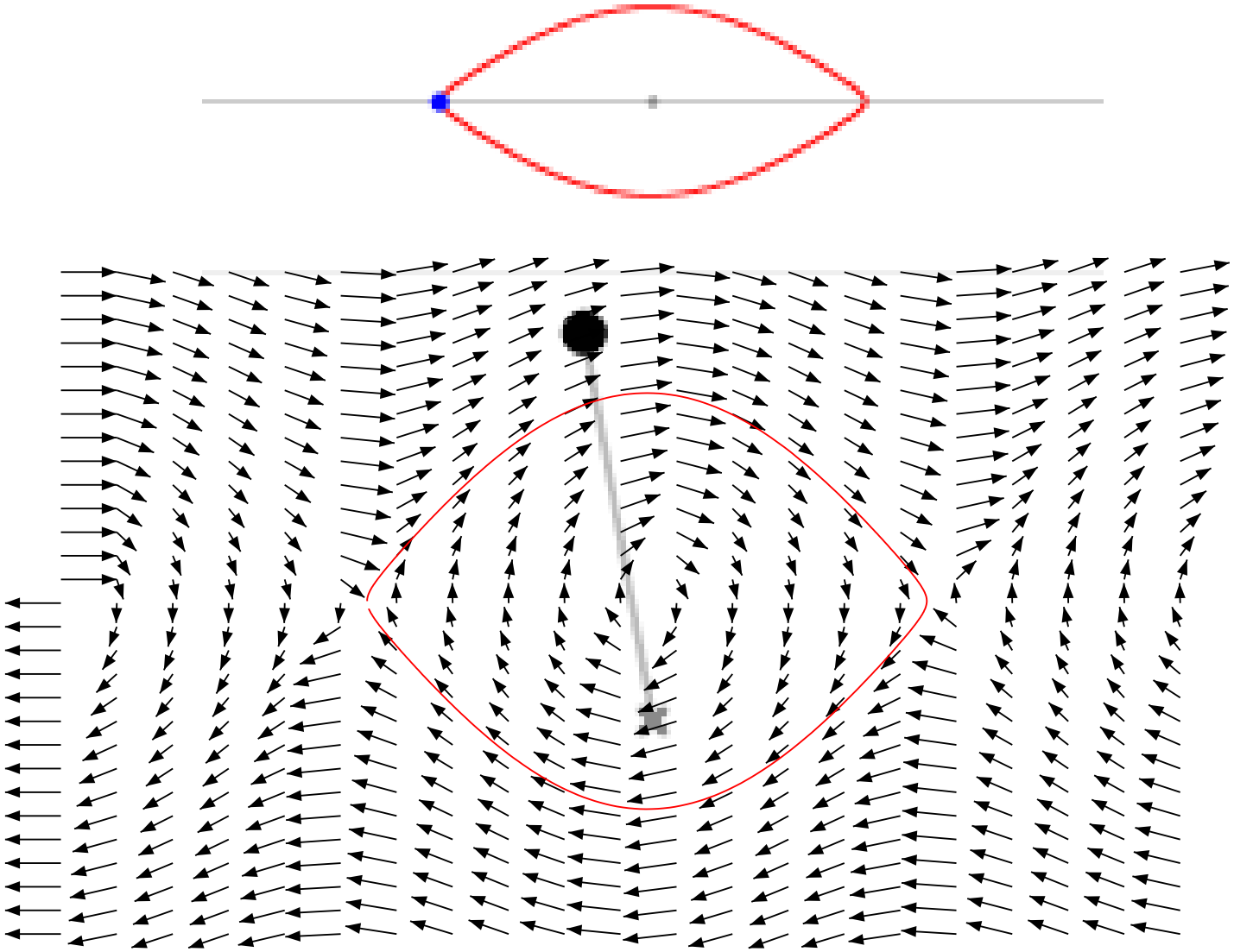


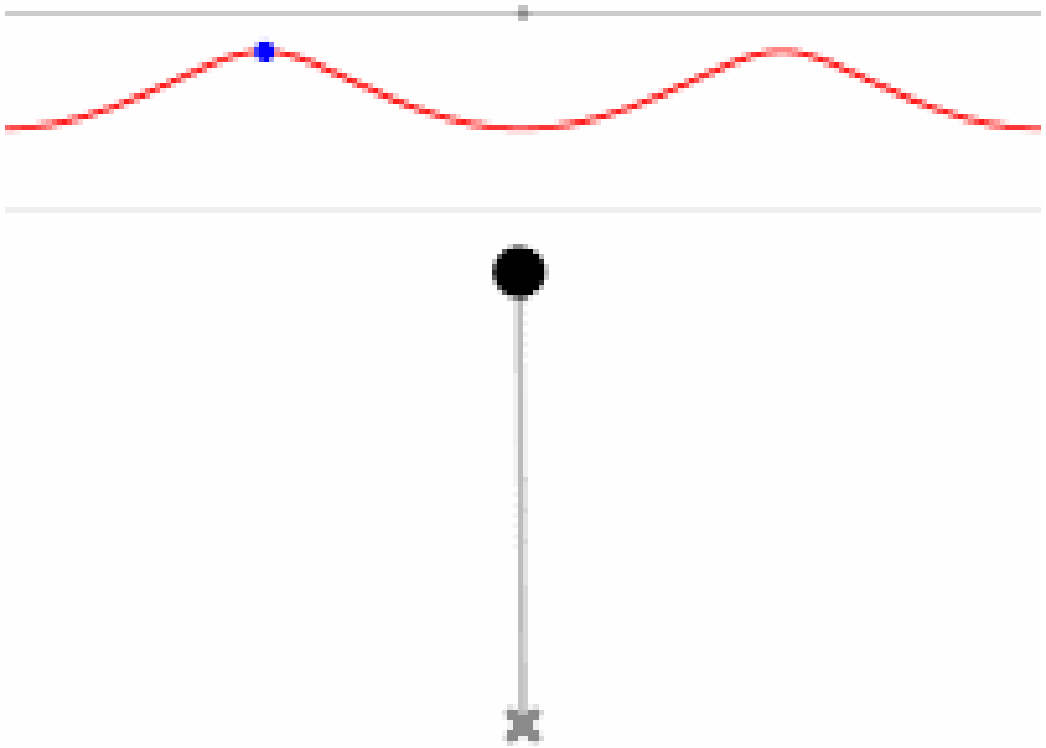


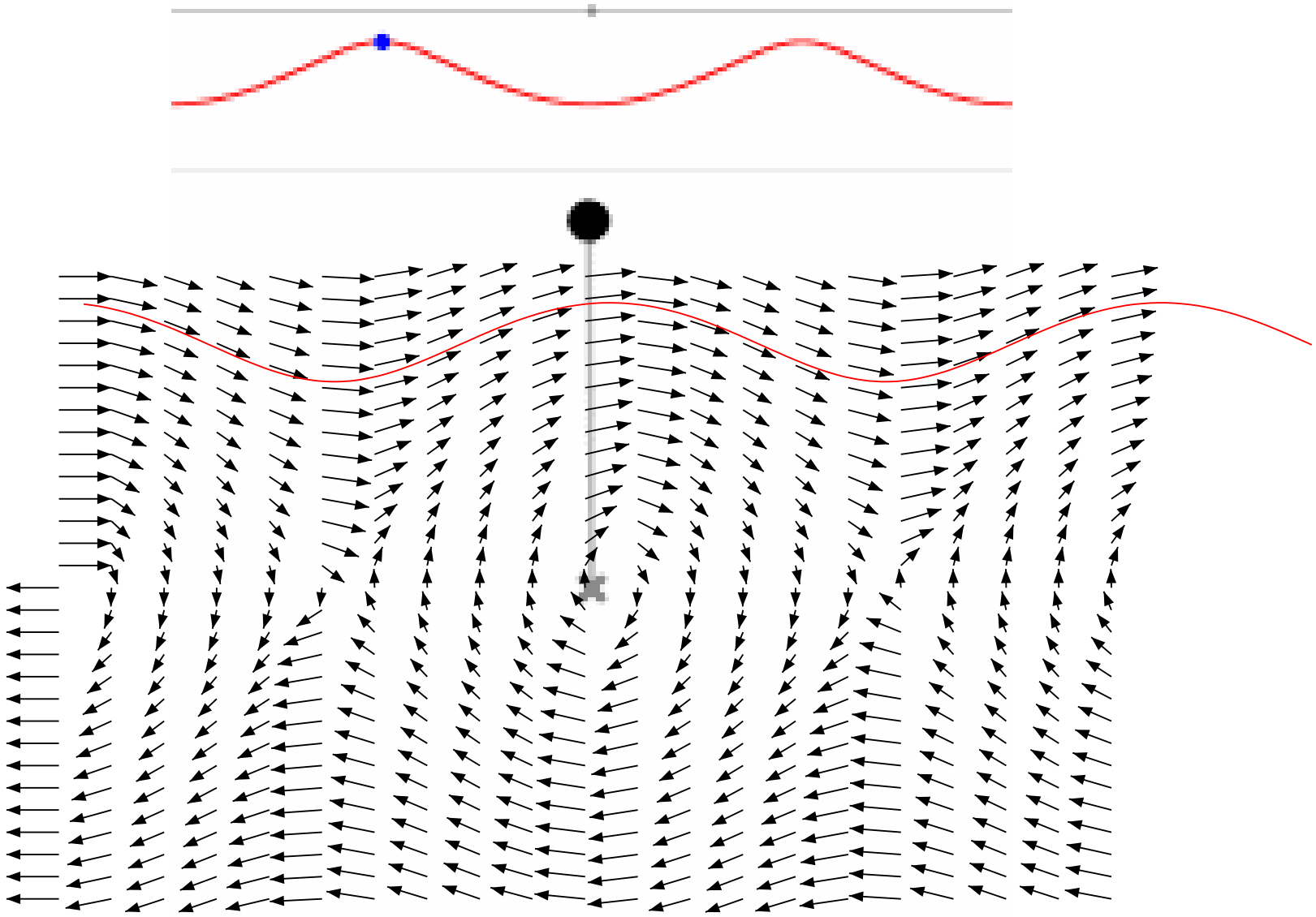








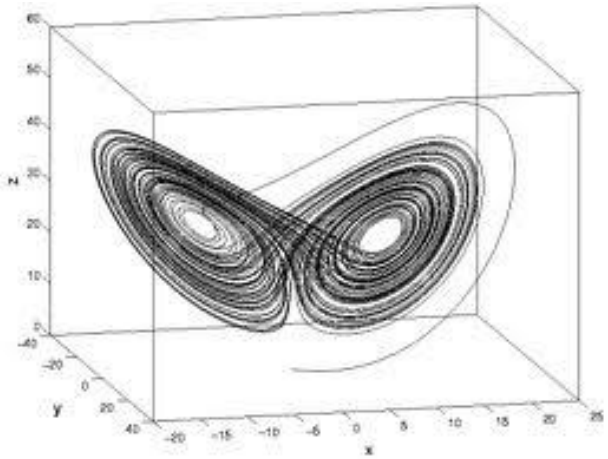




Flows and Chaos

In two dimensional systems you can have sources, sinks, saddle node points or limit cycles only.

In three dimensional systems like Lorenz system or Chua's circuit, much more complex behaviour is possible.

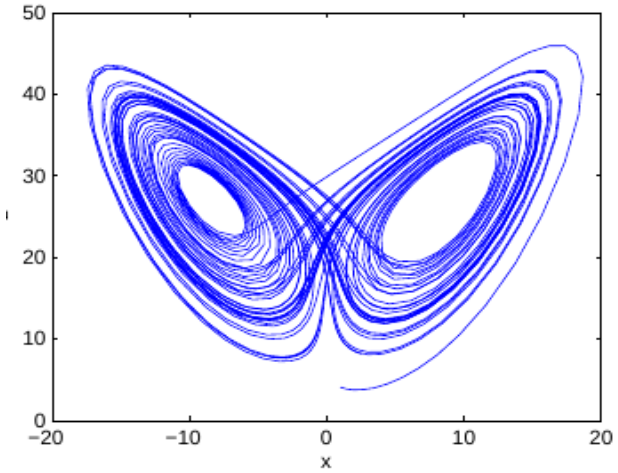


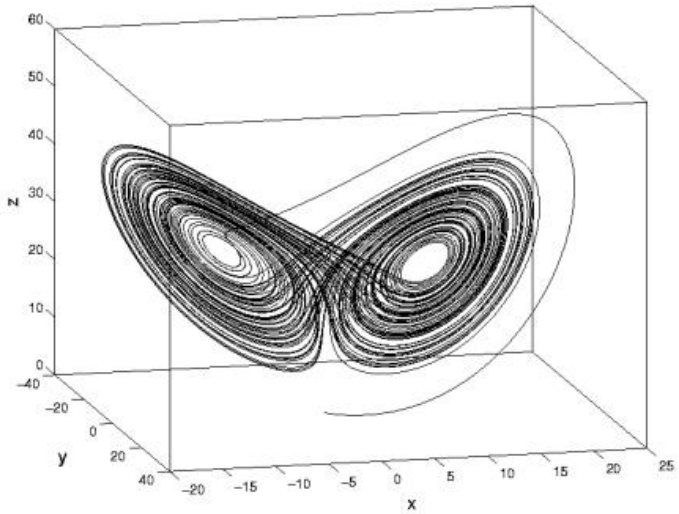
Drawing a flow diagram for a three dimensional system is difficult. Instead we can try to understand such systems using various analytic methods and by drawing trajectories.

Flows and Chaos

In Lorenz system we see two wings.

$$
\begin{aligned}
\frac{dx}{dt} &= -\sigma(x-y) \\
\frac{dy}{dt} &= -xz + rx - y \\
\frac{dz}{dt} &= xy - bz,
\end{aligned}
$$





Flows and Chaos

It is possible to show that this system has three equilibrium points.

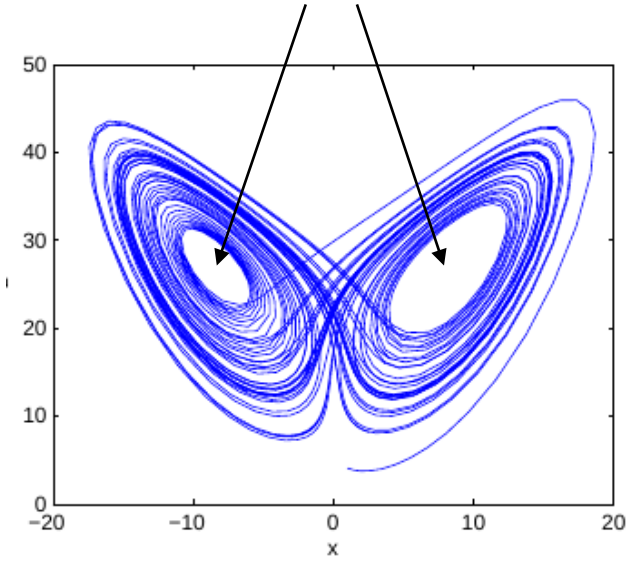
(0,0,0) is a trivial equilibrium point.

The other two are

$$\mathbf{c}_+ = [\sqrt{b(r-1)}, \sqrt{b(r-1)}, r-1],$$
$$\mathbf{c}_- = [-\sqrt{b(r-1)}, -\sqrt{b(r-1)}, r-1].$$

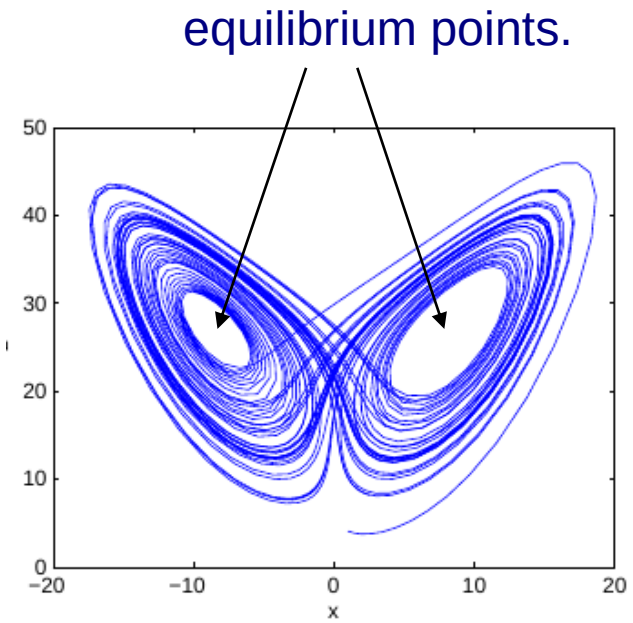
Under the parameter values that give the strange attractor, these two equilibrium points are unstable.

equilibrium points.

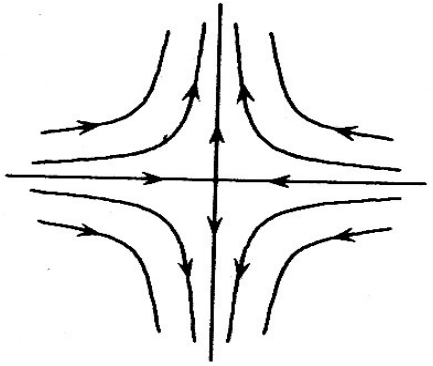


Flows and Chaos

It appears that, what is producing chaos is a competition between the two equilibrium points. They are attractive from some directions and repulsive from other sides.



A two dimensional example for such a point is the saddle node.

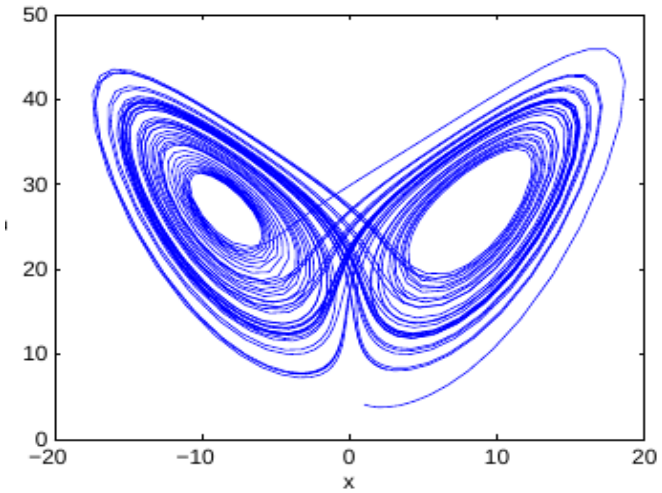


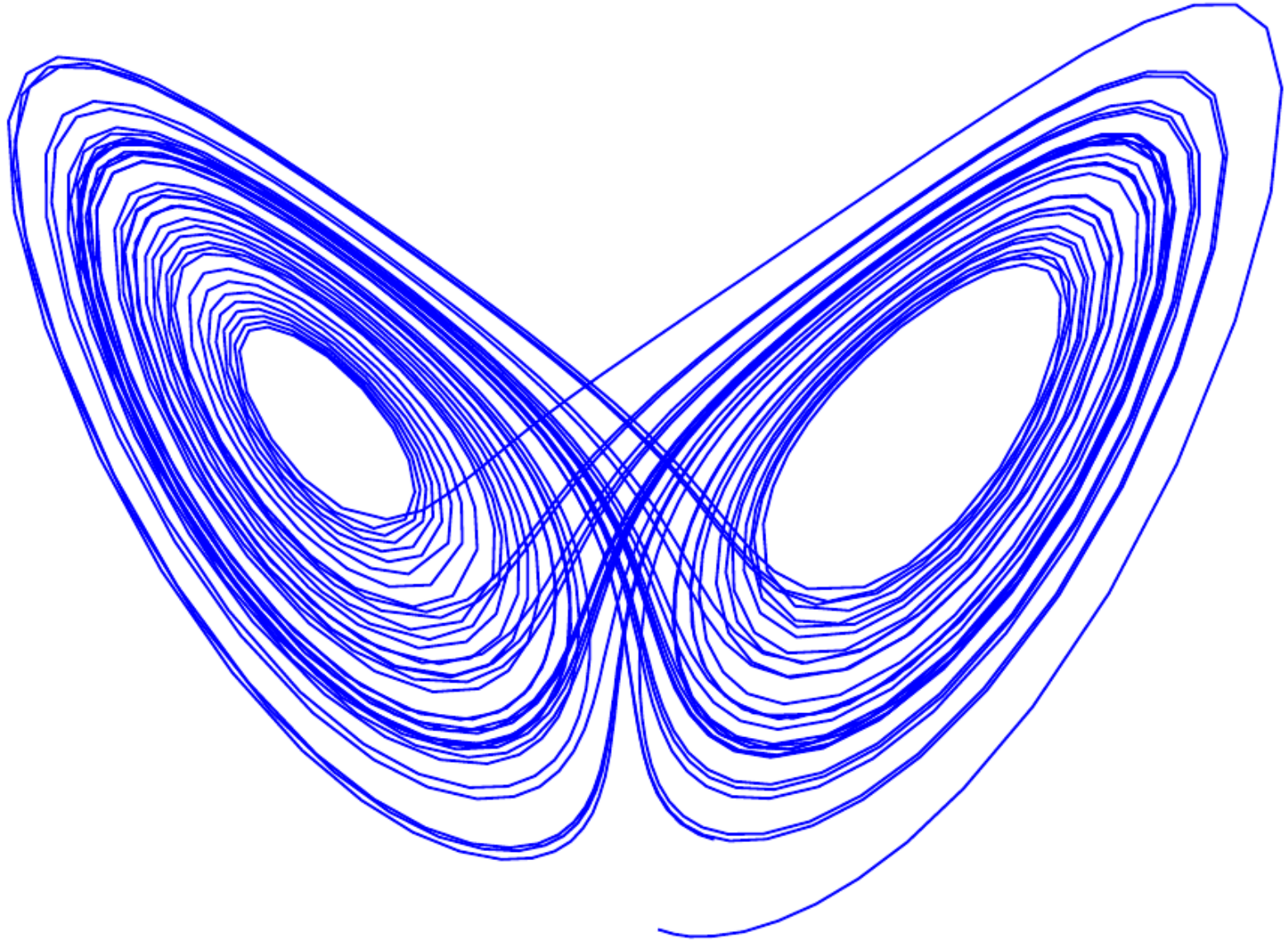
Flows and Chaos

Sensitivity to initial conditions is due to the very dense nature of the attractor.

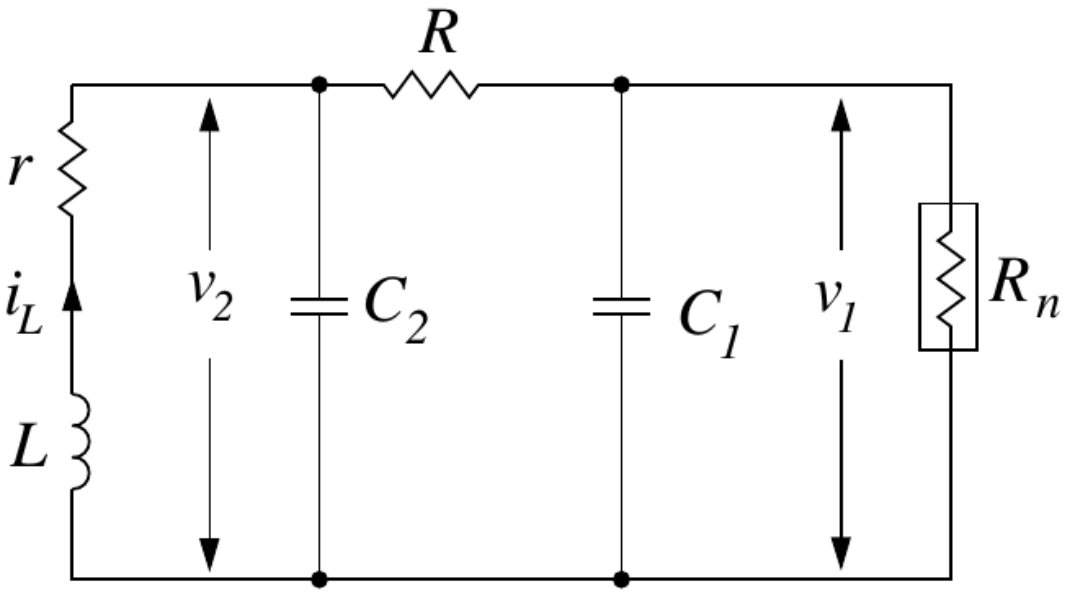
When we select one initial condition, the trajectory will fall on the attractor itself, or it will be attracted to one part of the attractor very fast.

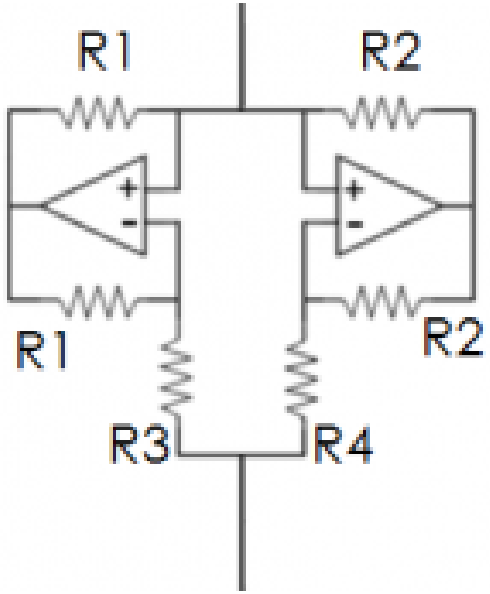
If we change the initial condition slightly, we may fall on or attracted to a different segment of the attractor.

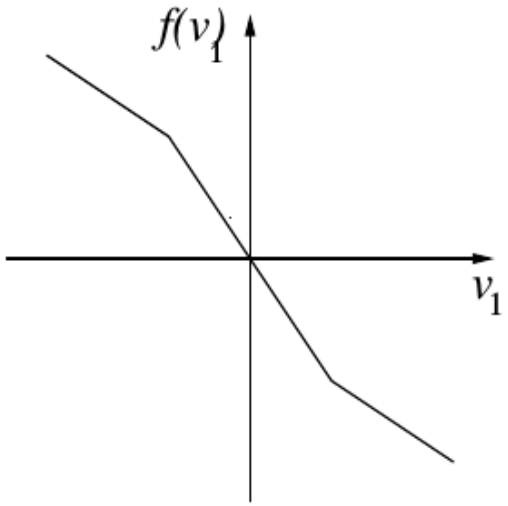




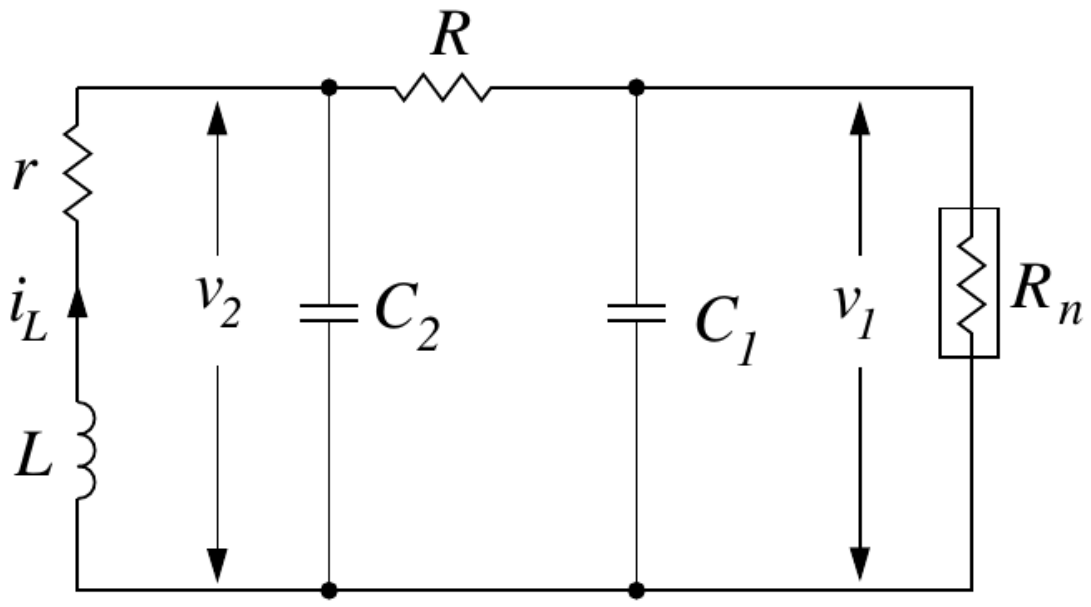
Chua's Circuit







Chua's Circuit

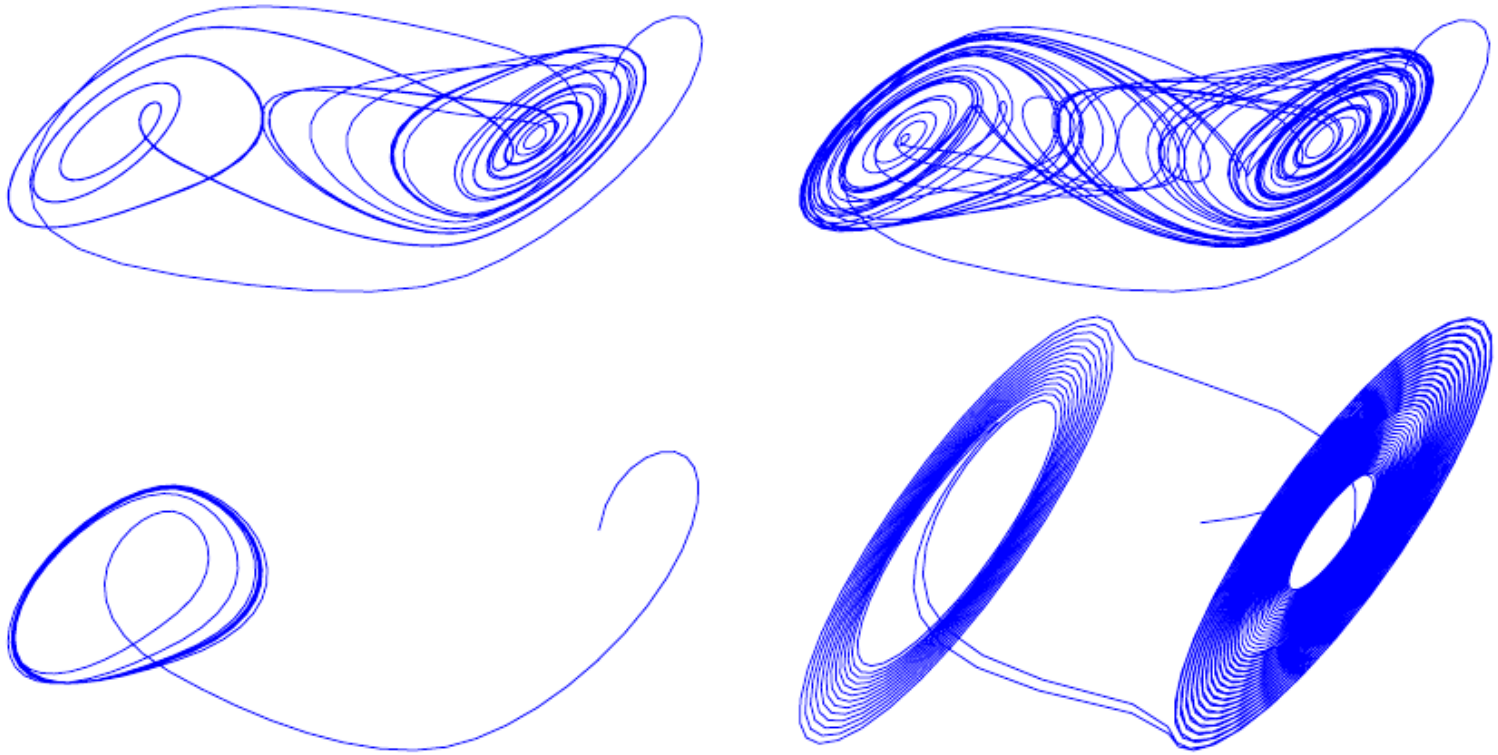


$$
\begin{aligned}
C_1 \frac{dv_1}{dt} &= \frac{1}{R}(v_2 - v_1) - f(v_1) \\
C_2 \frac{dv_2}{dt} &= \frac{1}{R}(v_1 - v_2) + i_L \\
L \frac{di_L}{dt} &= -v_2 - R_0 i_L
\end{aligned}
$$

$$
\begin{aligned}
C_1 \frac{dv_1}{dt} &= \frac{1}{R}(v_2 - v_1) - f(v_1) \\
C_2 \frac{dv_2}{dt} &= \frac{1}{R}(v_1 - v_2) + i_L \\
L \frac{di_L}{dt} &= -v_2 - R_0 i_L
\end{aligned}
$$

$$
x = \frac{v_1}{B_p} \qquad y = \frac{v_2}{B_p} \qquad z = \frac{i_L}{GB_p} \qquad \tau = \frac{tG}{C_2}
$$

$$
m_0 = \frac{G_a}{G} \qquad m_1 = \frac{G_b}{G} \qquad \alpha = \frac{C_2}{C_1} \qquad \beta = \frac{C_2}{LG^2} \qquad \gamma = \frac{R_0 C_2}{G}
$$

$$
\begin{aligned}
\frac{dx}{d\tau} &= \alpha\left(y - x - f(x)\right) \\
\frac{dy}{d\tau} &= x - y + z \\
\frac{dz}{d\tau} &= -\beta y - \gamma z
\end{aligned}
$$

$$
f(x) = m_1 x + \frac{1}{2}(m_0 - m_1)(|x+1| - |x-1|)
$$